ON THE ZEROS OF CERTAIN DIRICHLET SERIES

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1. The function $\zeta(s, a)$ is defined for $0 < a \leqslant 1$ by

$$\zeta(s, a) = \sum_{n=0}^{\infty} (n+a)^{-s}.$$

For $a = 1$, $\zeta(s, a) = \zeta(s)$, and for $a = \frac{1}{2}$, $\zeta(s, a) = (2^s - 1)\zeta(s)$; hence, for these two values of a, $\zeta(s, a)$ has no zeros in the half-plane $\sigma \geqslant 1$. We give in this note a simple proof that, for any other rational value of a, $\zeta(s, a)$ has an infinity of zeros for $\sigma > 1$. As regards irrational values of a, we show that the result holds also for any transcendental a. We are unable to attack the problem at all if a is an algebraic irrational number.

For a positive definite quadratic form

$$Q(x, y) = ax^2 + bxy + cy^2$$

with integer coefficients and a fundamental discriminant $d = b^2 - 4ac$, the Epstein zeta-function $\zeta(s, Q)$ is defined by

$$\zeta(s, Q) = \sum_{\substack{x, y = -\infty \\ x, y \neq 0, 0}}^{\infty} Q(x, y)^{-s}.$$

Potter and Titchmarsh† have recently proved that $\zeta(s, Q)$ has an infinity of zeros on $\sigma = \frac{1}{2}$, so that the analogue of Hardy's result holds. Using the same method as for $\zeta(s, a)$ with rational a, we prove that, if the class-number $h(d)$ is even, then $\zeta(s, Q)$ has an infinity of zeros for $\sigma > 1$‡. The condition $h(d)$ even is satisfied unless $d = -4$, -8, or $-p$, p prime§. If $h(d) = 1$, then

$$\zeta(s, Q) = c\zeta(s) \prod_p \left\{1 - \left(\frac{d}{p}\right) p^{-s}\right\}^{-1},$$

and so has no zeros for $\sigma > 1$. It can be proved that, if $h(d)$ is odd but different from 1, then again $\zeta(s, Q)$ has an infinity of zeros for $\sigma > 1$, but the proof is more complicated and will be given in a second note.

2. Let $a = l/k$, $k > 2$, $(l, k) = 1$, $0 < l < k$. If χ runs through the Dirichlet characters mod k, we have

$$\zeta(s, a) = \frac{k^s}{\phi(k)} \sum_{\chi} \bar{\chi}(l) L(s, \chi).$$

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† *Proc. London Math. Soc.* (2), 39 (1935), 372–384.

‡ The fact that not all zeros of $\zeta(s, Q)$ lie on $\sigma = \frac{1}{2}$ was discovered in the case $Q = x^2 + 5y^2$ by Potter and Titchmarsh (*loc. cit.*).

§ Hecke, *Theorie der algebraischen Zahlen* (1923), Satz 132.

We denote by χ_0 the principal character, and by χ_1 an arbitrary real non-principal character, which will be fixed throughout. For any prime p, we define

$$a(p) = \begin{cases} 1 & \text{if } \chi_1(p) = 0 \text{ or } 1, \\ i & \text{if } \chi_1(p) = -1; \end{cases}$$

and, for $n = p_1^{\nu_1} \dots p_r^{\nu_r}$, we define $a(n) = a(p_1)^{\nu_1} \dots a(p_r)^{\nu_r}$. We write

$$M(s, \chi) = \sum_{n=1}^{\infty} a(n)\chi(n)n^{-s} = \prod_p \{1 - a(p)\chi(p)p^{-s}\}^{-1},$$

$$Z(s) = \frac{k^s}{a(l)\phi(k)} \sum_\chi \bar{\chi}(l) M(s, \chi)$$

$$= \frac{k^s}{a(l)} \sum_{\substack{n=1 \\ n \equiv l \pmod k}}^{\infty} a(n)n^{-s}.$$

For $n \equiv l \pmod k$, we have $a(n) = \pm a(l)$; hence $Z(s)$ is real for $s > 1$, and it is easily seen that

$$\frac{1}{a(l)} \{M(s, \chi_0) + \chi_1(l) M(s, \chi_1)\} = 2\Re\left(\frac{M(s, \chi_0)}{a(l)}\right)$$

is also real for $s > 1$.

LEMMA 1. *For* $s > 1$,

$$\log M(s, \chi_0) = -\tfrac{1}{2}(1+i)\log(s-1) + O(1),$$

$$\log M(s, \chi_1) = -\tfrac{1}{2}(1-i)\log(s-1) + O(1),$$

and, for $\chi \neq \chi_0, \chi_1$,

$$\log M(s, \chi) = O(1).$$

Proof. We have

$$\log M(s, \chi) = \sum_p a(p)\chi(p)p^{-s} + O(1)$$

$$= \tfrac{1}{2}\sum_p \{1 + \chi_1(p)\}\chi(p)p^{-s} + \tfrac{1}{2}i\sum_p \{1 - \chi_1(p)\}\chi(p)p^{-s} + O(1)$$

$$= \tfrac{1}{2}(1+i)\log L(s, \chi) + \tfrac{1}{2}(1-i)\log L(s, \chi\chi_1) + O(1),$$

whence the result.

LEMMA 2. *For any* $\delta > 0$, $Z(s)$ *has a zero in* $1 < s < 1+\delta$.

Proof. By the last part of Lemma 1,

$$Z(s) = \frac{2k^s}{\phi(k)} \Re\left(\frac{1}{a(l)} M(s, \chi_0)\right) + O(1)$$

for $s > 1$. By the first part of Lemma 1, we can, for any $H > 0$, find s_1 and

s_2 such that $1<s_1<s_2<1+\delta$ and

$$\Re\left(\frac{1}{a(l)}M(s_1, \chi_0)\right) > H,$$

$$\Re\left(\frac{1}{a(l)}M(s_2, \chi_0)\right) < -H.$$

Then, if H is sufficiently large,

$$Z(s_1) > 0 > Z(s_2).$$

Thus there is a value of s with $1<s<1+\delta$ for which $Z(s)=0$. This proves the lemma.

THEOREM 1. *If a is rational and $a \neq \frac{1}{2}$ or 1, $\zeta(s, a)$ has an infinity of zeros for $\sigma > 1$.*

Proof. By Kronecker's theorem*, given $\epsilon > 0$ and $P > 2$, there exists a real number τ such that $|p^{-i\tau}-a(p)| < \epsilon$ for all $p \leqslant P$. Hence, for any $\epsilon > 0$ and $\delta > 0$, there is a τ such that

$$|\zeta(s+\tau i, l/k)-a(l)\,Z(s)| < \epsilon$$

for $\sigma > 1+\delta$.

Let $s_1 > 1$ be a zero of $Z(s)$. For any $\eta > 0$ there exists an η_1 with $0<\eta_1<\eta$, $\eta_1 < s_1-1$, such that $Z(s) \neq 0$ for $|s-s_1| = \eta_1$. Take

$$\epsilon = \min_{|s-s_1|=\eta_1} |Z(s)|,$$

and $\delta < s_1-\eta_1-1$; then the conditions of Rouché's theorem are satisfied on $|s-s_1| = \eta_1$. Hence $\zeta(s, l/k)$ has a zero in the circle $|s-s_1-\tau i| < \eta$. This proves the theorem.

3. THEOREM 2. *For transcendental a, $\zeta(s, a)$ has an infinity of zeros for $\sigma > 1$.*

Proof. The numbers $\log(n+a)$ are linearly independent. For a relation

$$\sum_{n=0}^{N} c_n \log(n+a) = 0$$

with integers c_n not all zero implies that

$$\prod_{n=0}^{N} (a+n)^{c_n} = 1,$$

and this cannot be an identity in a since it is not satisfied by $a = -n$ if $c_n \neq 0$.

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* Titchmarsh, *The zeta-function of Riemann* (1930), 11.

For any $\delta > 0$ we choose m such that

$$\sum_0^m (n+a)^{-1-\delta} > \sum_{m+1}^{\infty} (n+a)^{-1-\delta},$$

and let $a(n) = 1$ for $n \leqslant m$ and -1 for $n > m$. Then

$$Z(s) = \sum_0^{\infty} a(n)(n+a)^{-s}$$

has a zero for $1 < s < 1+\delta$. Applying Kronecker's theorem as before, we find that $\zeta(s, a)$ has a zero with $1 < \sigma < 1+\delta$.

4. Let Q be a positive definite quadratic form with a fundamental discriminant d for which $h(d)$ is even, and so $d \leqslant -15$. It is well known* that the various non-equivalent quadratic forms of discriminant d correspond (1, 1) to the classes $\mathfrak{K}$ of ideals in the quadratic field $\mathrm{P}(\sqrt{d})$ in such a way that the number of representations of a positive integer n by the quadratic form is precisely twice the number of integer ideals of norm n in the corresponding class. Hence, if $\mathfrak{K}_1$ is the class corresponding to Q,

$$\zeta(s, Q) = 2 \sum_{\mathfrak{a} \text{ in } \mathfrak{K}_1} N\mathfrak{a}^{-s}.$$

Let χ run through the characters of the group of ideal classes, and let χ_0 denote the principal character. Then

$$\zeta(s, Q) = \frac{2}{h(d)} \sum_{\chi} \bar{\chi}(\mathfrak{K}_1) L(s, \chi),$$

where

$$L(s, \chi) = \sum_{\mathfrak{a}} \chi(\mathfrak{a}) N\mathfrak{a}^{-s} = \prod_{\mathfrak{p}} \{1 - \chi(\mathfrak{p}) N\mathfrak{p}^{-s}\}^{-1},$$

for $\sigma > 1$, where $\mathfrak{a}$ runs through all integer ideals other than 0, and $\mathfrak{p}$ through all prime ideals.

It is known that, for $\chi \neq \chi_0$, $L(s, \chi)$ is regular at $s = 1$, and $L(1, \chi) \neq 0$, also that $L(s, \chi_0)$ has a simple pole at $s = 1$.

Since the order of the class-group is even, there exists† a real non-principal character χ_1.

THEOREM 3. *$\zeta(s, Q)$ has an infinity of zeros for $\sigma > 1$.*

The proof is identical with that of § 3, except for one point in connection with the application of Kronecker's theorem. The numbers $N\mathfrak{p}$ are not

* Hecke, *Theorie der algebraischen Zahlen* (1923), § 53.

† The number of characters, being the order of the class-group, is even; hence the number of real characters is even.

all different, but the same $N\mathfrak{p}$ can arise only from two different $\mathfrak{p}$'s, and the product $\mathfrak{p}_1\mathfrak{p}_2$ of two such $\mathfrak{p}$'s is a rational prime. Hence $\chi_1(\mathfrak{p}_1)\chi_1(\mathfrak{p}_2) = 1$, *i.e.* $a(\mathfrak{p}_1) = a(\mathfrak{p}_2)$. Thus the application of Kronecker's theorem to the logarithms of the set* consisting of the different values assumed by $N\mathfrak{p}$ is sufficient.

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NOTE ON THE UNIFORMIZATION OF HYPERELLIPTIC CURVES

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It was shown by E. T. Whittaker‡ that the uniformizing variable of the curve

$$u^2 = f(z) \equiv \prod_{m=1}^{2n+2} (z - e_m)$$

$$\equiv z^{2n+2} + \sum_{m=1}^{2n+2} (-1)^m p_m z^{2n-m+2},$$

is given by the differential equation

$$\tfrac{1}{2}\{t, z\} = \tfrac{3}{16}\left\{\sum_{m=1}^{2n+2} \frac{1}{(z-e_m)^2} + \frac{\psi(z)}{f(z)}\right\},$$

where $\{t, z\}$ is the Schwarzian derivative, and

$$\psi(z) \equiv -(2n+2)z^{2n} + 2np_1 z^{2n-1} + \sum_{r=1}^{2n-1} c_r z^{2n-r-1},$$

the coefficients c_r depending on $e_1, e_2, \ldots$ in such a manner that the group of linear transformations of t is fuchsian.

Adopting the hypothesis that each c_r is a polynomial in $e_1, e_2, \ldots$, J. M. Whittaker§ has shown that

$$c_r = (-1)^r (2n-r+1)(2n-r)p_{r+1}/(2n+1),$$

so that

$$\psi(z) = -f''(z)/(2n+1).$$

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* The set consists of all primes with $(d/p) = 0$ or 1, and the squares of all primes with $(d/p) = -1$, and the logarithms of these numbers are linearly independent.

† Received 28 January, 1936; read 20 February, 1936.

‡ E. T. Whittaker, *Phil. Trans. Royal Soc.* (A), 192 (1898), 1–32.

§ J. M. Whittaker, *Journal London Math. Soc.*, 5 (1930), 150–154.